

EVPK Simulator

User Manual

A practical guide to the web app at igem.ardabakici.com for pharmacokinetic (PK) modeling (1c/2c/3c), regimen comparison, and pharmacodynamic (PD) links (bacteria CFU kill and PMM2 activity rescue). Screenshots and button-by-button instructions are included.

Source code: github.com/Aman-Sunesh/EV-PK-Simulator

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1 Introduction

The PB-PK Simulator is a browser-based toolkit for:

- Fitting 1-, 2-, or 3-compartment macro-exponential PK models to CSVs.
- Simulating dosing routes and regimens (IV bolus, short infusion, oral; repeat dosing).
- Comparing multiple regimens on one plot with key KPIs.
- Performing PD analyses driven by PK outputs: bacteria CFU dynamics and PMM2 activity rescue.
- Generating one-click PDF reports that capture parameters, selection/demotion rationale, plots, and data-hygiene warnings.

Direct access. The hosted app is available at igem.ardabakici.com.

Run locally (optional). If you prefer a local instance, follow the backend/frontend steps in Section 2.

2 Installation (Local Development)

2.1 Prerequisites

Python 3.10+ and Node.js 18+ recommended.

2.2 Backend

```
# From repo root
pip install -r requirements.txt

cd backend
python3 -m venv venv
source venv/bin/activate

# Run FastAPI dev server on http://0.0.0.0:8000
uvicorn main:app --reload --host 0.0.0.0 --port 8000
```

2.3 Frontend

```
cd frontend
npm install
npm start
```

The frontend dev server typically opens at <http://localhost:3000> and proxies API calls to the backend (<http://localhost:8000>).

3 App Overview (Home)

The home page presents the main modules as large tiles: One-/Two-/Three-Compartment Models, PD Analysis from Data, and Compare Regimens.



Figure 1: Home gateway tiles to PK and PD tools.

Click a tile to open that module. Navigation links at the top-left return you to the Home at any time.

4 Three-Compartment Model (PK)

This screen is representative of the single- and two-compartment pages; fields differ only by the number of macro terms.

The image shows the "Three-Compartment Model" interface. At the top left, there is a link "← Home". The title "PB-PK Simulator" is in blue. Below it, the "Preset" is set to "EV (IV, mouse)" with a help icon and a description: "EV defaults preload $t_{1/2} \approx 0.7-3.5$ h ($kel \approx 0.2-1.0$ h⁻¹) and faster distribution." The main title "Three-Compartment Model" is in bold. Below it, the "Species" is "Mus musculus" and the "Dose" is "100 mg". The "Parameter source" section has two options: "Estimate from data (fit)" and "Simulate: Injection Route Explorers & Dosing Regimens" (selected). A note says: "Simulate: pick a route (Route), define the regimen (Dosing), or explore sliders (What-If). Then click Simulate." The "Form" is set to "Macro (A, α , B, β , C, γ)". Below this, there are input fields for parameters: A (1/L): 0.6, α (1/h): 1.2, B (1/L): 0.3, β (1/h): 0.3, C (1/L): 0.1, and γ (1/h): 0.06. The "Simulate" panel at the bottom has a "Route" button (selected) and a "Dosing" button. There are "Clear plots" and "Simulate" buttons. The "Route" is set to "IV bolus". The "t_end (h)" is 24 and "dt (h)" is 0.1. There is a checkbox for "Semilog Y" which is unchecked.

Figure 2: Three-Compartment model — parameter inputs and simulate panel.

4.1 Header & Preset

- **Preset:** quick-fill convenience values (e.g., species-specific defaults).
- Inline tip text reminds IV bolus half-life ranges and distribution expectations.

4.2 Inputs

- **Species:** free text (for labeling).
- **Dose (mg):** total amount for a single administration.
- **Parameter source:**
 - *Estimate from data (fit)* — upload a CSV to fit macro parameters.
 - *Simulate: Injection Route Explorers & Dosing Regimens* — directly enter macro parameters and simulate.
- **Form:** macro parameters. For 3c you enter $A, \alpha, B, \beta, C, \gamma$; for 2c omit (C, γ) ; for 1c you can use (V_d, k) or (A, α) form.

4.3 Simulate Panel (bottom of page)

- **Route:** choose the input route (IV bolus; IV infusion (short, overlap/non-overlap); oral with first-order absorption).
- **Dosing:** define dose events: bolus at time t , infusion with start/duration, oral repeats, or a programmatic repeat every τ .
- t_{end} (**h**): simulation horizon. dt (**h**): time step.
- **Semilog Y:** toggles a log-scale concentration axis.
- **Buttons:** **Clear plots** (reset) and **Simulate** (run and plot).

Output. The plot shows $C(t)$; above it KPIs such as **Cmax**, **Tmax**, and **AUC** ($0-t_{\text{end}}$) are summarized.

5 Simulate: Route/Dosing Builder (All Models)

This screen exposes a plan-builder when you click **Route** or **Dosing** (or open the dedicated *Simulate* page). You can stack multiple inputs and remove them.

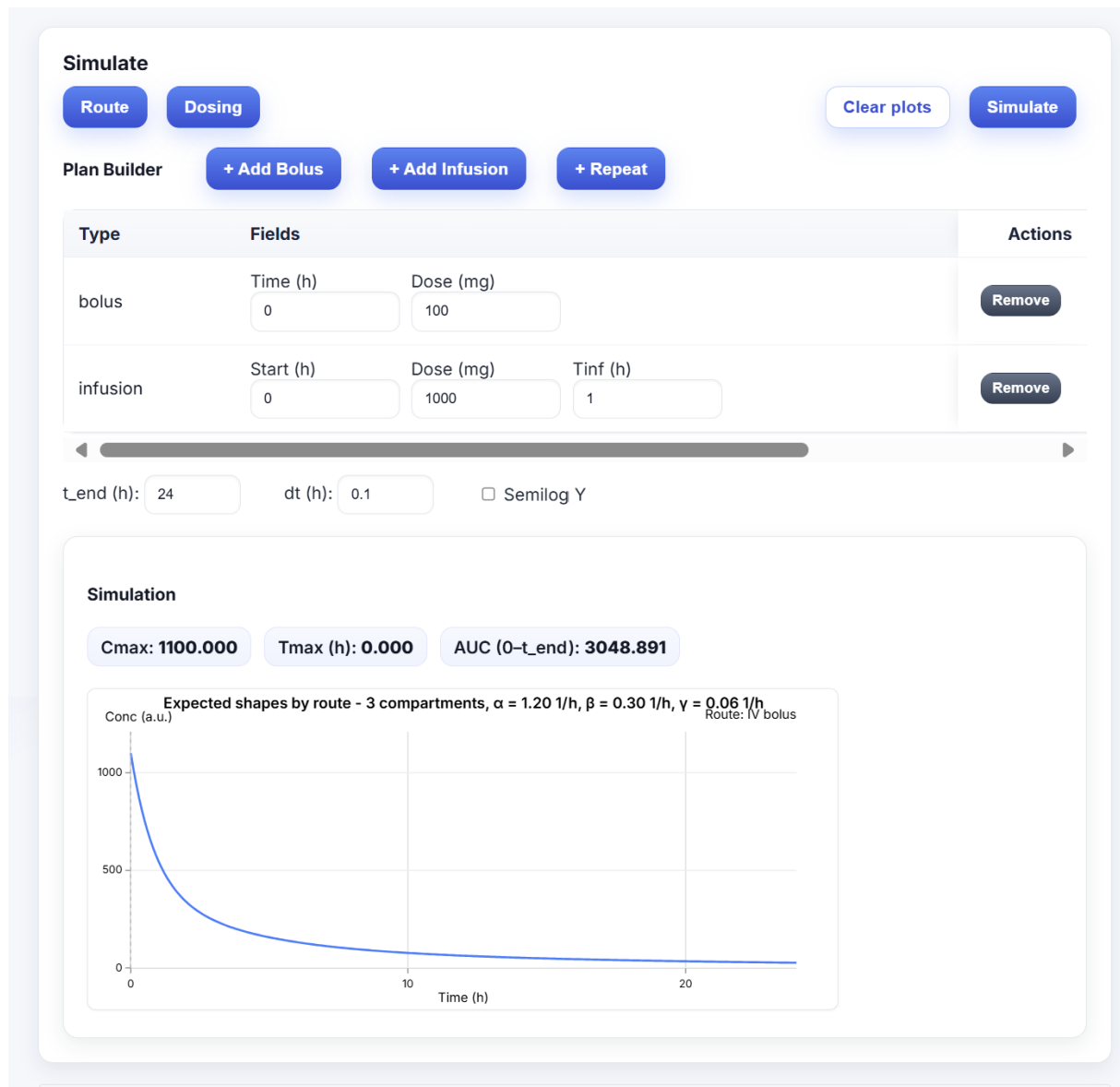


Figure 3: Plan Builder: add bolus & infusion events, then simulate. KPIs appear above the plot.

Controls:

- **+ Add Bolus:** Time (h), Dose (mg). Creates an impulse input.
- **+ Add Infusion:** Start (h), Dose (mg), T_{inf} (h). Adds a zero-order input lasting T_{inf} .
- **+ Repeat:** repeat a prior pattern every τ hours (steady-state experiments).
- **Actions → Remove:** delete a row.
- Bottom controls: t_{end} , dt , Semilog Y toggle, and **Simulate**.

Readouts:

- **Cmax, Tmax, AUC:** auto-computed for the shown window.
- Plot: linear (or semi-log) concentration profile.

6 Compare Regimens (Side-by-side PK)

Define multiple scenarios across rows, then render them on a common timeline.

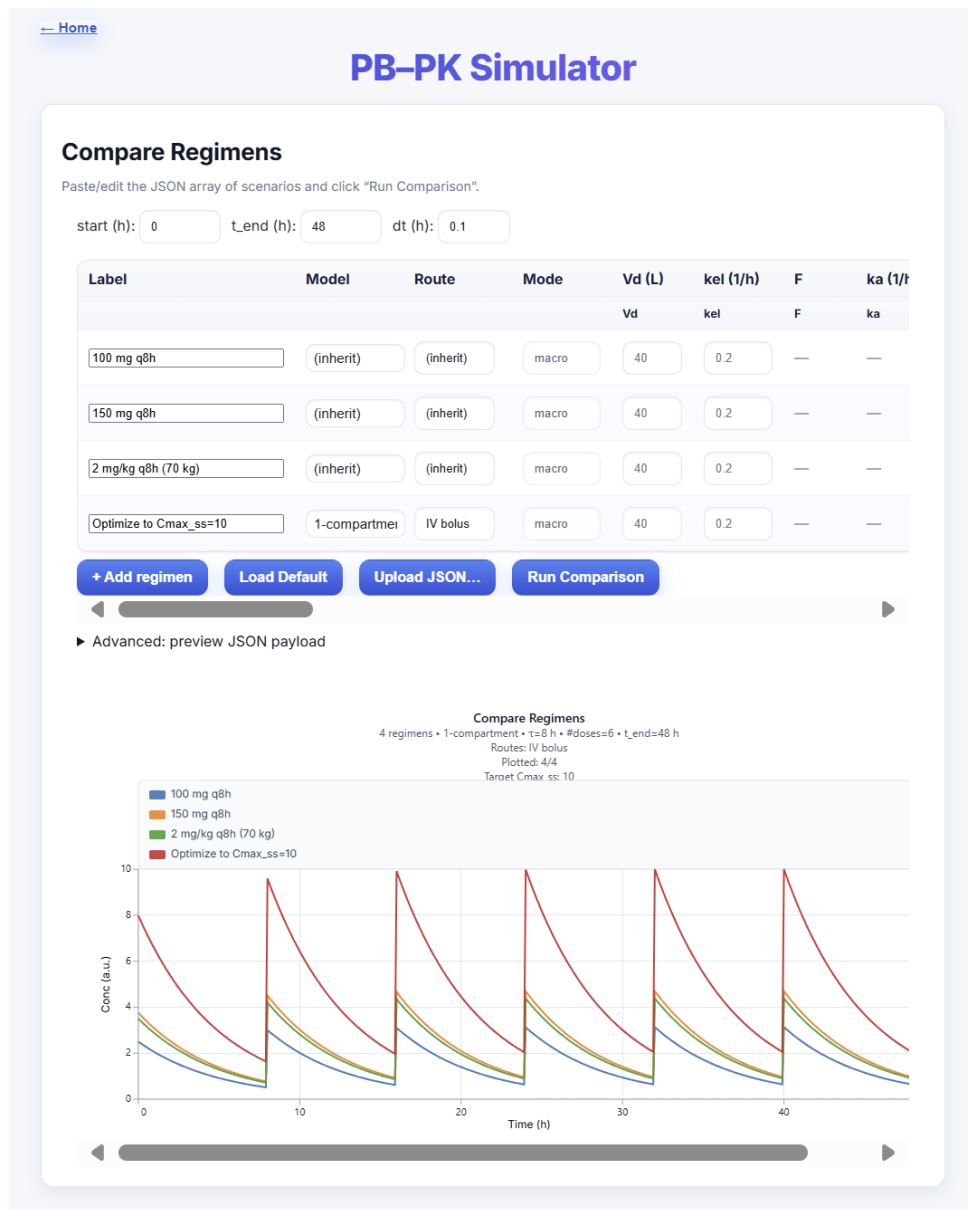


Figure 4: Compare up to many regimens with a shared time base. Advanced: preview JSON payload.

Top timeline fields (global to the comparison):

- **start (h)**, **t_end (h)**, **dt (h)**: simulation window and resolution.

Regimen table (one row per scenario):

- **Label**: legend name.
- **Model**: inherit (use the current page model) or override.
- **Route/Mode**: IV bolus or infusion; macro vs micro entry (usually macro).
- **Vd, kel, F, ka**: visible when relevant for 1c macro/micro forms.

Buttons:

- **+ Add regimen:** append a new row.
- **Load Default:** populate a few example rows.
- **Upload JSON...:** load a saved comparison payload.
- **Run Comparison:** compute and render.

Advanced (accordion):

- *Preview JSON payload:* inspect the exact request that the UI sends.

Output: Overlaid $C(t)$ curves with legend and optional target annotations (e.g., *Target $C_{max,ss} = 10$*).

7 PD Analysis from Data

This module accepts raw *concentration–time* CSVs (from your lab or a PK simulation) and computes PD outcomes using built-in link models.

PD Analysis from Data

PD Analysis from Raw Data
Upload your concentration–time data and perform pharmacodynamic analysis directly without requiring PK model fitting. The raw uploaded data will be used as concentration inputs for PD modeling.

Data Input (Choose One) Clear & Switch

Option 1: Load Example Study

Load Example: — select study —

OR

Option 2: Upload File ✓

Drag & drop a CSV/Excel file here, or click to browse
Expected format: columns for Time, Concentration, and optionally Subject

Data Preview

subject	species	dose	time	concentration
S1	Homo sapiens	100	0	7.531013
S1	Homo sapiens	100	0.1	6.118837
S1	Homo sapiens	100	0.25	5.646263
S1	Homo sapiens	100	0.5	4.712244
S1	Homo sapiens	100	0.75	4.165597

Pharmacodynamic Analysis

PD Model: Bacteria CFU Dynamics

Bacteria CFU Dynamics Parameters

Initial CFU (CFU0): 1000000

Max Kill Rate (k_max): 1 1/h

EC50 for Kill: 1 mg/L

Hill Coefficient: 1

Growth Rate (k_grow): 0 1/h

Run PD Analysis Clear plots

Figure 5: PD Analysis from Data: drag & drop CSVs, preview the table, then configure the PD model.

7.1 Data Input

- **Load Example:** pick a built-in small dataset (sanity check).
- **Drag & drop / click to browse:** upload your CSV/Excel. Expected columns include time, concentration, and optionally subject. The preview table displays the parsed data.

7.2 Pharmacodynamic Analysis Panel

- **PD Model** selector:
 - *Bacteria CFU Dynamics* (logistic growth with concentration-dependent kill).
 - *PMM2 Activity Rescue* ($E_{\max}$ /Hill link).
- **Buttons:** Run PD Analysis, Clear plots.

Outputs (after a run):

- Summary badges (e.g., *Initial CFU*, *Final CFU*, *Log Kill*).
- Time-series plots (e.g., $\log_{10}$ CFU vs time; %Activity vs time).
- **Download PD Report:** export a PDF with inputs, fitted/used PD parameters, and plots.

8 Standalone PD Workbench (Quick Run)

For quick bacterial-kill or PMM2-rescue explorations, you can open the PD workbench directly and enter PD parameters (without uploading data).

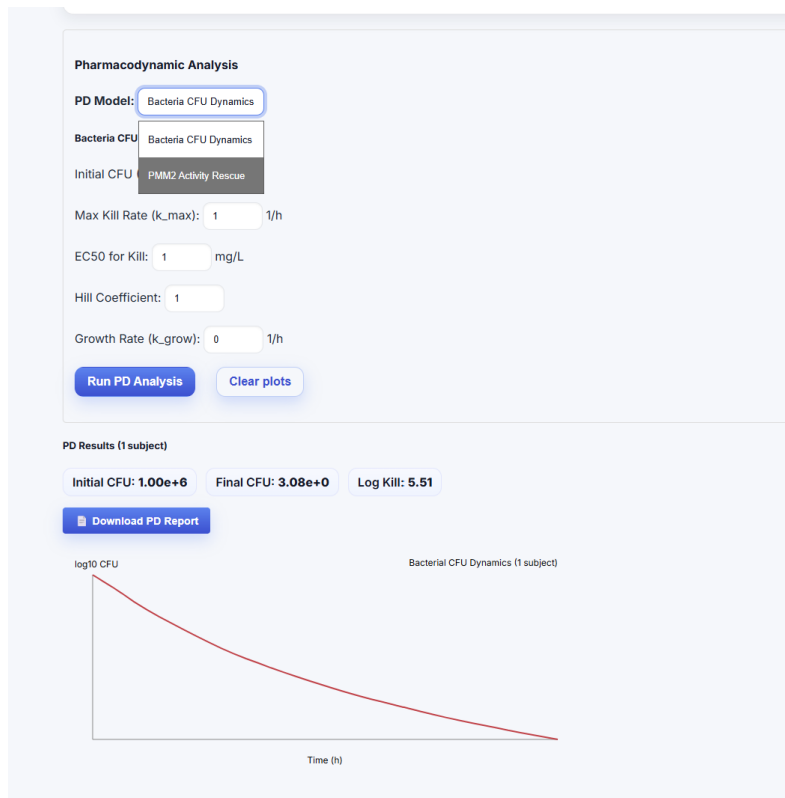


Figure 6: Bacteria CFU Dynamics: enter N_0 , $k_{\max}$, EC50, Hill h , and growth rate; then run.

8.1 Bacteria CFU Dynamics

Inputs:

- **Initial CFU** (CFU_0).
- **Max Kill Rate** ($k_{\max}$) [1/h].
- **EC50 for Kill** [mg/L].
- **Hill Coefficient** h .
- **Growth Rate** (k_{grow}) [1/h].

Outputs:

- *Initial CFU, Final CFU, Log Kill.*
- Plot of $\log_{10}$ CFU vs time.
- **Download PD Report.**

8.2 PMM2 Activity Rescue

Inputs (typical):

- E_{basal} , E_{max} , **EC50**, **Hill** h .

Output: %Activity time-course, suitable for target-attainment checks (e.g., maintain $\geq \theta\%$).

9 Fit From Data (PK)

When the **Parameter source** is set to *Estimate from data (fit)*, upload a CSV on the PK page to estimate macro parameters per subject. The *Data Input* screen (for example on the 3c page) looks like this:

The screenshot shows the 'PB-PK Simulator' interface. At the top, there's a 'Home' link and a title 'PB-PK Simulator'. Below the title, it says 'Preset: EV (IV, mouse)' and a note: 'EV defaults preload $t_{1/2}=0.7-3.5$ h ($k_{el}=0.2-1.0$ h⁻¹) and faster distribution.' The main section is titled 'Three-Compartment Model'. Under 'Data Input (Choose One)', there are two options: 'Option 1: Load Example Study' and 'Option 2: Upload File'. Option 1 includes a 'Load Example:' dropdown menu. Option 2 features a large dashed box with the text 'Drag & drop a CSV/Excel file here, or click to browse'. Below these options, there are fields for 'Species:' (set to 'Mus musculus') and 'Dose:' (set to '100 mg'). At the bottom, the 'Parameter source:' is set to 'Estimate from data (fit)' (indicated by a selected radio button), with an alternative option 'Simulate: Injection Route Explorers & Dosing Regimens'. A note below says 'Analyze: upload a dataset and fit parameters.' and a checkbox 'Allow demotion (3→2→1) by AICc & rate separation' is checked.

Figure 7: PK fitting data input: choose example or upload; species/dose are for labeling and scaling.

9.1 Model Selection & Demotion (under the hood)

The app fits 1c/2c/3c macro-exponential models and compares them with AICc. Identifiability guards demote 3→2 or 2→1 if:

- ΔAICc favors the simpler model (e.g., > 2),
- the minimum relative rate separation $\min_{i < j} \frac{|r_i - r_j|}{\max(r_i, r_j)}$ is too small (merged phases),
- the slowest tail AUC fraction is below a threshold (negligible tail).

Badges on the report explain the choice (selected model, ΔAICc , n , demotion reason, and any preprocessing warnings).

10 Mathematical Notes (Macro Forms)

For reference, single-dose closed forms used throughout (superposition handles repeats; steady state uses geometric series):

One compartment (1c).

$$C(t) = \frac{D}{V_d} e^{-k \Delta t}, \quad \Delta t \equiv t - t_D \geq 0.$$

Two compartments (2c).

$$C(t) = D \left(A e^{-\alpha \Delta t} + B e^{-\beta \Delta t} \right), \quad \alpha \neq \beta.$$

Three compartments (3c).

$$C(t) = D \left(A e^{-\alpha \Delta t} + B e^{-\beta \Delta t} + C e^{-\gamma \Delta t} \right), \quad \alpha > \beta > \gamma > 0.$$

Selection metric (linear-space residuals).

$$\text{AICc} = 2k - 2 \ln \hat{L} + \frac{2k(k+1)}{n-k-1}, \quad \ln \hat{L} = -\frac{1}{2\sigma^2} \sum_i (y_i - \hat{y}_i)^2 + \text{const.}$$

Identifiability scores.

$$\text{sep}_{\text{rel}} = \min_{i < j} \frac{|r_i - r_j|}{\max(r_i, r_j)}, \quad r \in \{\alpha, \beta, \gamma\}, \quad \phi_{\text{tail}, 3c} = \frac{C/\gamma}{A/\alpha + B/\beta + C/\gamma}.$$

11 CSV Schema, Units, & Hygiene

- Core columns (typical): `species`, `subject`, `time`, `concentration`, `dose`.
- Optional per-row dose: `dose_mg_per_kg`, `weight_kg` (enables subject-specific dosing).
- Units: time in hours; concentration in mg/L; dose in mg (unless stated).

Preprocessing policy:

- Drop non-positive concentrations (warning shown).
- Mean-aggregate duplicate (`subject`, `time`) rows (warning).
- Apply a tiny floor only for log plots to avoid $\log(0)$; outliers are otherwise kept.

12 Exporting Results

Most pages include **Download PDF report**. Exports typically contain:

- Parameter tables (with confidence intervals when available).
- Plots: linear/semi-log $C(t)$, residuals, or PD time-series.
- Badges: n , $AICc/\Delta AICc$, selected model, demotion reason, and hygiene warnings.

13 Troubleshooting

Frontend

- **Blank plot/controls unresponsive:** refresh; check browser console for CORS errors if running locally.
- **“npm start” fails:** ensure Node 18+; remove `node_modules` & `package-lock.json` then `npm install`.

Backend

- **Port 8000 in use:** add `--port 8001` and update frontend proxy.
- **CSV rejected:** confirm required columns (columns "time" and "concentration" are required); ensure decimal points (not commas).

PK/PD specifics

- **Model demoted unexpectedly:** examine rate separation and tail-AUC badges; noisy data may not justify extra phases.
- **Noisy or sparse data:** prefer $AICc$ -driven parsimony; try the Compare Regimens or Simulate route to explore design space.
- **PD run flatlines:** verify non-zero $k_{\max}$, reasonable EC_{50} and Hill; check units of input concentration.

14 Figures (Gallery)

[← Home](#)

PB-PK Simulator

Preset: EV (IV, mouse) ? EV defaults preload $t_{1/2} \approx 0.7-3.5$ h ($kel \approx 0.2-1.0$ h⁻¹) and faster distribution.

Three-Compartment Model

Species: Mus musculus

Dose: 100 mg

Parameter source: ☐ Estimate from data (fit) ☒ Simulate: Injection Route Explorers & Dosing Regimens

Simulate: pick a route (Route), define the regimen (Dosing), or explore sliders (What-If). Then click Simulate.

Form: Macro (A, α , B, β , C, γ)

A (1/L): 0.6 α (1/h): 1.2 B (1/L): 0.3 β (1/h): 0.3 C (1/L): 0.1 γ (1/h): 0.06

Simulate

Route: IV bolus

t_end (h): 24 dt (h): 0.1 ☐ Semilog Y

Figure 8: Three-Compartment Model (top).

[← Home](#)

PB-PK Simulator

Preset: EV (IV, mouse)

? EV defaults preload $t_{1/2} \approx 0.7\text{--}3.5\text{ h}$ ($k_{el} \approx 0.2\text{--}1.0\text{ h}^{-1}$) and faster distribution.

Three-Compartment Model

Data Input (Choose One)

Option 1: Load Example Study

Load Example: — select study —

OR

Option 2: Upload File

📁 Drag & drop a CSV/Excel file here, or **click to browse**

Species: Mus musculus

Dose: 100 mg

Parameter source:

☒ Estimate from data (fit)

☐ Simulate: Injection Route Explorers & Dosing Regimens

Analyze: upload a dataset and fit parameters.

☒ Allow demotion (3→2→1) by AICc & rate separation

Figure 9: PK Fit: Data input & options.

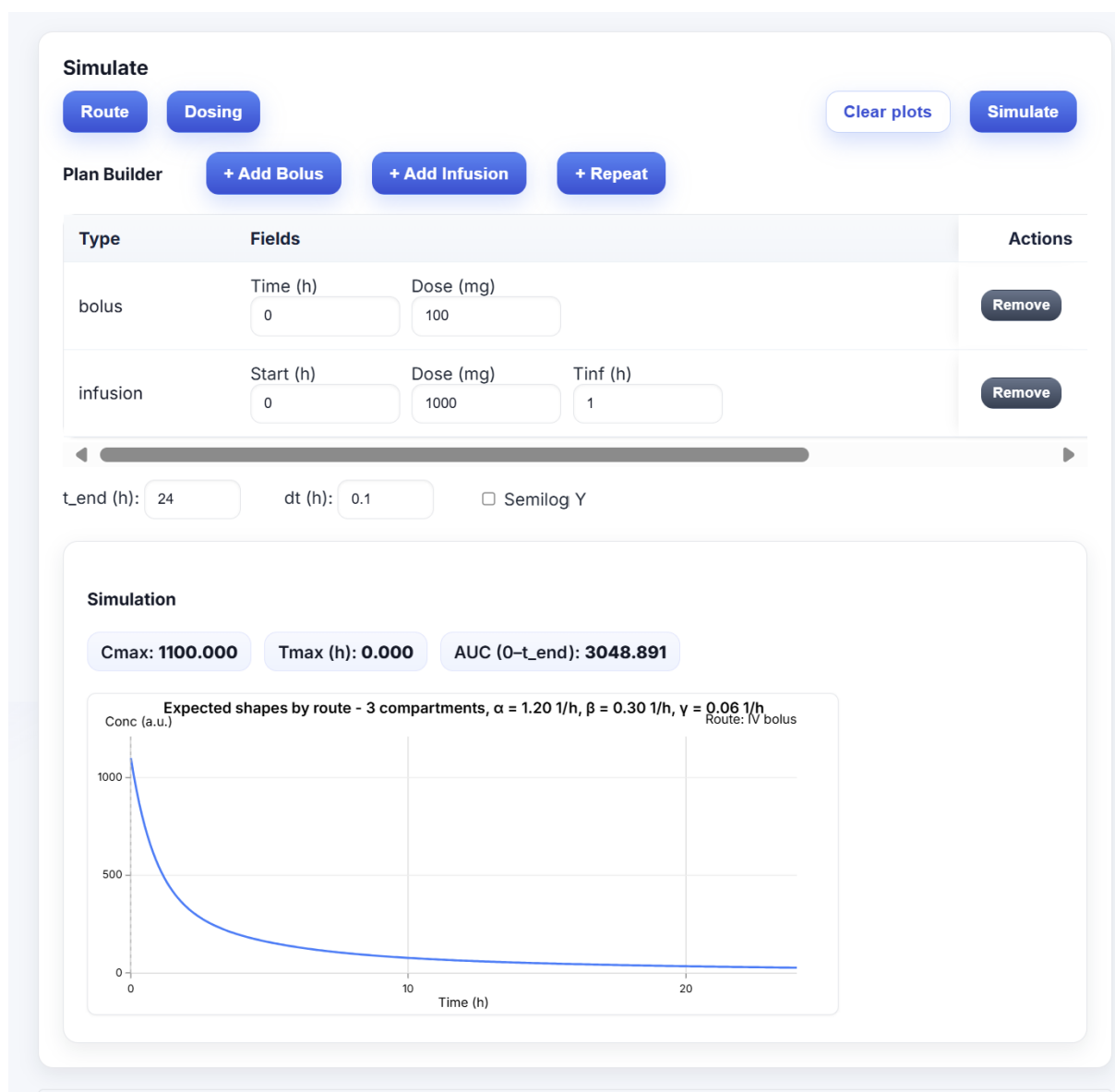


Figure 10: Simulate: Plan Builder and KPIs.

PB-PK Simulator

Compare Regimens

Paste/edit the JSON array of scenarios and click "Run Comparison".

start (h): t_end (h): dt (h):

Label	Model	Route	Mode	Vd (L)	kel (1/h)	F	ka (1/h)
				Vd	kel	F	ka
100 mg q8h	(inherit)	(inherit)	macro	40	0.2	—	—
150 mg q8h	(inherit)	(inherit)	macro	40	0.2	—	—
2 mg/kg q8h (70 kg)	(inherit)	(inherit)	macro	40	0.2	—	—
Optimize to Cmax_ss=10	1-compartment	IV bolus	macro	40	0.2	—	—

+ Add regimen

Load Default

Upload JSON...

Run Comparison

► Advanced: preview JSON payload

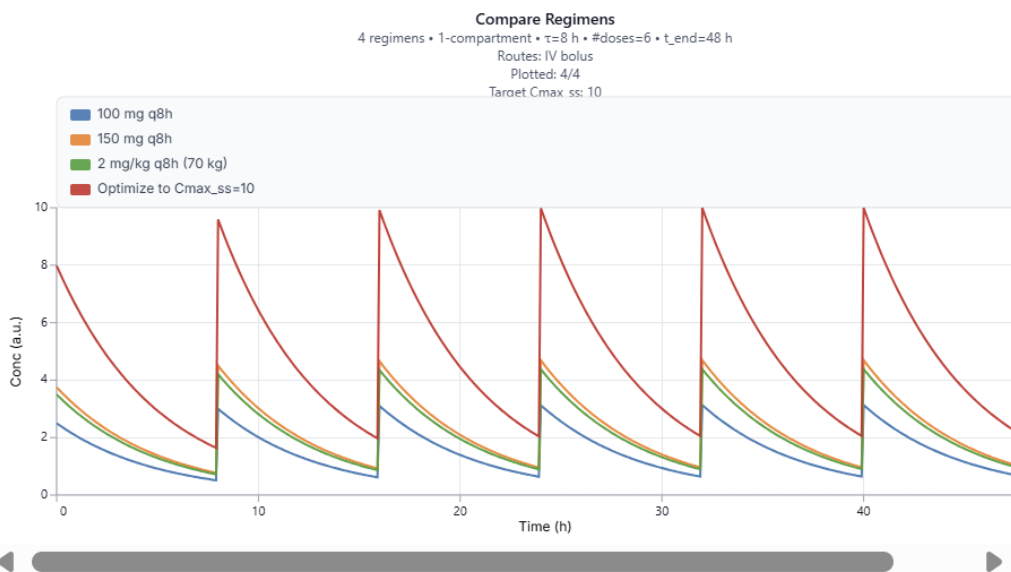


Figure 11: Compare Regimens: multiple scenarios on one plot.

PD Analysis from Data

PD Analysis from Raw Data

Upload your concentration-time data and perform pharmacodynamic analysis directly without requiring PK model fitting. The raw uploaded data will be used as concentration inputs for PD modeling.

Data Input (Choose One)

Clear & Switch

Option 1: Load Example Study

Load Example:

— select study —

OR

Option 2: Upload File ✓

📁 Drag & drop a CSV/Excel file here, or **click to browse**

Expected format: columns for Time, Concentration, and optionally Subject

Data Preview

subject	species	dose	time	concentration
S1	Homo sapiens	100	0	7.531013
S1	Homo sapiens	100	0.1	6.118837
S1	Homo sapiens	100	0.25	5.646263
S1	Homo sapiens	100	0.5	4.712244
S1	Homo sapiens	100	0.75	4.165597

Pharmacodynamic Analysis

PD Model:

Bacteria CFU Dynamics

Bacteria CFU Dynamics Parameters

Initial CFU (CFU0):

1000000

Max Kill Rate (k_max):

1

 1/h

EC50 for Kill:

1

 mg/L

Hill Coefficient:

1

Growth Rate (k_grow):

0

 1/h

Run PD Analysis

Clear plots

Figure 12: PD Analysis from Data.

16

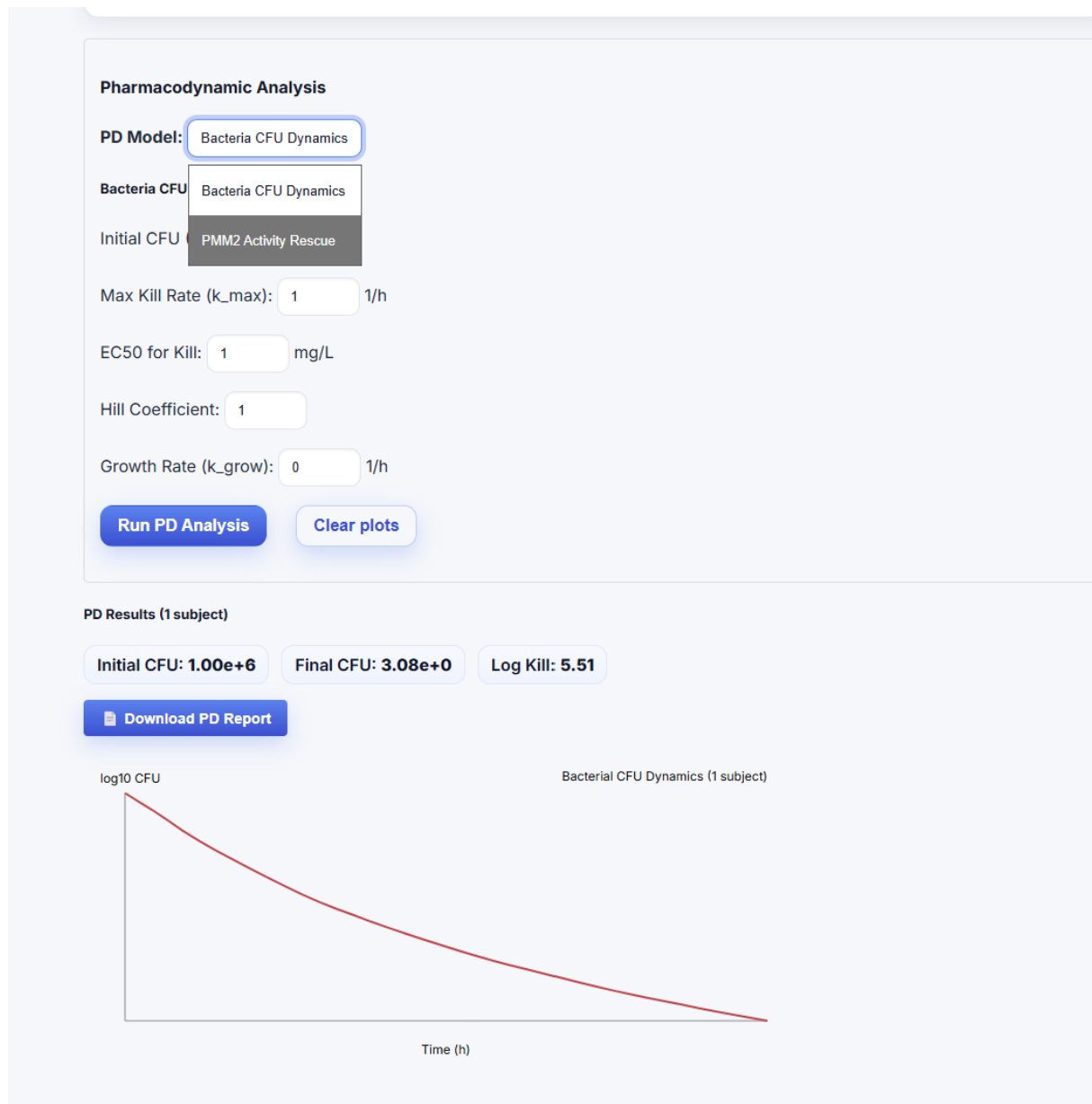


Figure 13: Standalone PD Workbench: Bacteria CFU Dynamics.

15 References

- Internal project docs on datasets, selection/demotion logic, and route formulas (see comment citations at top of this $\text{\LaTeX}$ file).
- App (hosted): igem.ardabakici.com